Voltage Control Practices and Tools Used for System Voltage Control of PJM

Jianzhong Tong, *Senior Member, IEEE*, David W. Souder, Christopher Pilong, Mingye Zhang, Qinglai Guo, Hongbin Sun, *Member, IEEE*, Boming Zhang, *Fellow, IEEE*

***Abstract* —PJM has developed a pilot program of on-line (real time) voltage control in which a system wide Optimal Voltage Control (OVC) demonstration system has been implemented and run on-line parallel with PJM existing system at PJM control center since Jan. 2010. The pilot program was for nine months.**

The OVC system consists of Data I/O Management, Control Strategy Calculation, Operation Log Management and Control Process Dispatch. It is interfaced with PJM EMS system, automatically receives real-time snapshot from PJM EMS periodically, and calculates optimal voltage schedules for each generator, LTC and switchable shunt in the PJM system.

The practices, experiences and results of the pilot program are presented in this panel paper. Significant potential benefits can be achieved with OVC techniques in both aspects of security and economy:

***(1)* average reduction of transmission system loss is about 1.19%, which means energy savings of more than 220 million kWh/year and monetary savings of more than $17 million/year;**

***(2)* Average increase of system VAR reserve is about 1.08%;**

***(3)* Improvement on system security for pre-contingency and post-contingency is significant, including improvement on these security indices in voltage limits violations, and voltage stability limits violations.**

***Index Terms* — Automatic Voltage Control, Optimal Voltage Schedule, PJM Real Time Operations.**

I. Introduction

PJM transmission system has the generation capacity of 164,000 MW, and the peak load of 145,000 MW. With the increase of PJM power grids scale, voltage control and Var management become more and more challenging. So far each TO in the PJM system determines their own area voltage schedule based on their individual assessments. If a TO has no voltage schedule for its area, PJM defines a default one. This suboptimal approach is no longer suitable for the higher security and economy requirements of the PJM system operations.

PJM has planned to develop an application of System-Wide Optimal Voltage Control (OVC) for the PJM system operations. A collaborative project on the development of the OVC application system between PJM and Tsinghua University has been carried out since 2008, which has the following three phases:

- Phase I: OVC study on PJM system.
- Phase II: A pilot program for online implementation and evaluation of OVC at PJM control center.
- Phase III: OVC system fully productized at PJM control center.

Hierarchical voltage control architecture [1-7] was developed for PJM OVC system, which is based on the concepts of zone division and pilot buses. As Phase I, the OVC study on PJM system has been completed as of March, 2009. The study was done by off-line testing with eight typical real-time snapshots obtained from the solutions of state estimator in the PJM's energy management system (EMS), The results of evaluation show that potential benefits are significant with OVC in both aspects of security and economy, including system losses reduction, system MVAR reserve increase, and improvement on system security for pre-contingency and post-contingency [8].

In phase II, PJM has developed a pilot program of on-line (real time) voltage control in which a system wide Optimal Voltage Control (OVC) demonstration system has been implemented and run on-line parallel with PJM existing system at PJM control center for nine months. A commercial AVC system [7] which was developed by Tsinghua University and has been applied to more than a dozen control centers in China was used for the development of the PJM OVC system. The OVC system consists of Data I/O Management, Control Strategy Calculation, Operation Log Management and Control Process Dispatch. It is interfaced with PJM EMS system, automatically receiving real-time snapshot from PJM EMS on a periodical base, and calculating optimal voltage schedules for each generator, LTC and switchable shunt in the PJM system. The practices, experiences and results of the pilot program are presented in this panel paper.

II. Overview of OVC System

The online OVC system was developed to import real-time snapshot from PJM's EMS automatically, output a power flow solution and optimal voltage schedule, manage and dispatch all OVC system's modules to run in a correct order, and record software running information into logs. In the calculations of optimization, the reactive outputs of all running generators, LTC and switchable shunts inside of PJM system were selected as control variables, and all necessary voltage/VAR constraints inside of PJM system were

Jianzhong Tong (e-mail: tongji@pjm.com), David Souder, Chris Pilong are with the PJM interconnection L.L.C., PA19403, USA.

Mingye Zhang, Qinglai Guo, Hongbin Sun and Boming Zhang are all with the Department of Electrical Engineering, State Key Laboratory of Power Systems, Tsinghua University, Beijing 100084, China.

978-1-4577-1002-5/11/$26.00 ©2011 IEEE

considered.

The overview architecture of implemented OVC system is shown in Fig. 1. The OVC system is interfaced with PJM EMS to receive State Estimation (SE) solutions with real time network models and a contingency list from Security Analysis (SA) periodically. With those two inputs, OVC system carries out the optimization of control strategy calculation and outputs optimal voltage schedule for each generator, LTC and switchable shunt within the PJM system.

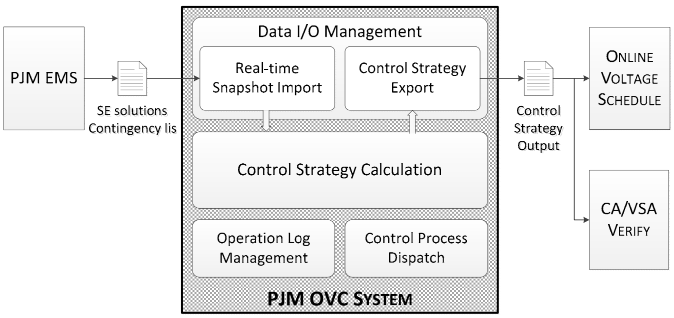

Fig 1 Architecture of OVC demonstration system

The software modules of OVC system include Data I/O Management, Control Strategy Calculation Optimization), Log Management and Control Process Dispatch.

A. Data I/O Management

1) Data Input

The PJM EMS creates a snapshot periodically, including all the input files

- Power flow data file in PSS/E format.
- Monitor list file with limits of monitored voltage included.
- Contingency list file in PSS/E format with more than 5,500 contingencies.

2) Data Output

For each real-time snapshot, when the calculation of optimal voltage control was completed, all the result files will be outputted to a special directory on OVC server. The result files include:

- Optimal voltage control result file in PSS/E format.
- Optimal voltage schedule data file in PI database format.
- Summary file with some statistics of voltage, transmission losses and reactive power reserves of both original snapshot and the optimal voltage control result.

B. Control Strategy Calculation

Once real-time snapshot data was imported to OVC system successfully, and the interval between the last complete snapshot and the new snapshot is longer than the setting value, 45 minutes for example, AVC system will start the optimal voltage control calculation automatically with control strategy calculation module.

OVC system is a centralized control system, which can coordinate the actions of all voltage/VAR control devices across the whole power grid. The control strategy calculation module in it is responsible for the optimal control decision-making for the entire power grid.

Logically the control strategy calculation module consists of two levels, i.e. the secondary voltage control (SVC) and Tertiary Voltage Control (TVC). The logical hierarchical structure was shown in Fig. 2. TVC offers the optimal set-points of the pilot bus voltages to SVC, and then the latter coordinates all generators to keep the voltages at the pilot buses on their optimal reference values.

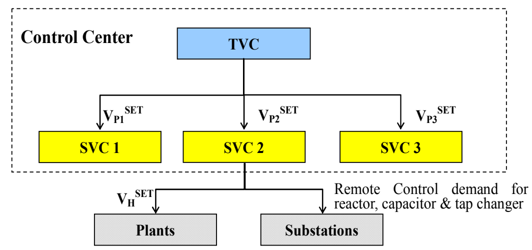

Fig 2 Hierarchical structure of control strategy calculation module

Control strategy calculation adopts the reactive optimal power flow (ROPF) with the goal of system loss minimization to obtain the optimal set-points of the pilot bus voltages. In ROPF calculation model, reactive outputs of all generators inside of PJM system are considered as control variables. Security constraints are considered in high dimensions as well, which include all voltage constraints and VAR constraints inside the PJM system. Since control strategy presented from OVC should meet requirements of Contingency Analysis (CA), CA results were used in the voltage constraints on critical buses. If the two objectives are conflict to each other, control strategy calculation will make a trade-off between them by readjusting the voltage constraints on critical buses.

C. Log Management

OVC system was supposed to run periodically without manual intervention. The Log management system was designed for any trouble shooting. Two kinds of log files were created and saved by OVC automatically. One was the overview of the control process; another was the detail information for debugging.

D. Control Process

An independent module is running to dispatch the whole control process from data import to online schedule export and to make a right response if there is any mistake, shown in Fig 3. Furthermore, to keep OVC system running normally for 24 hours without any exceptional quit, a daemon process is adopted to monitor the operation status of OVC system.

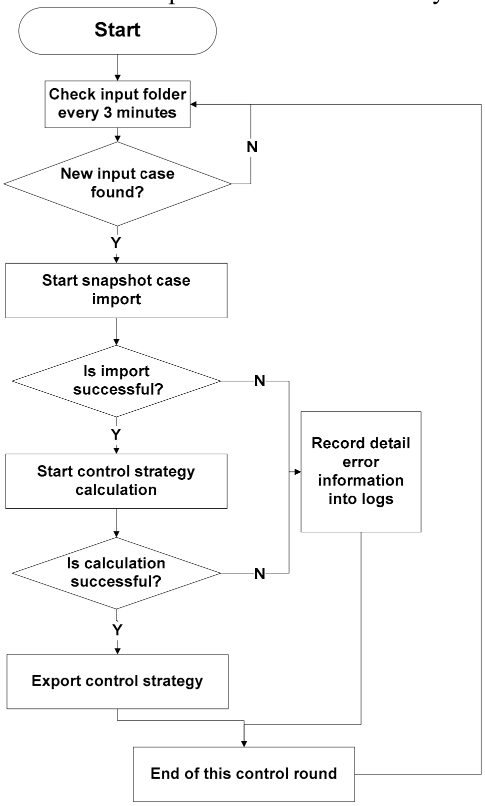

Fig 3 Control dispatch diagram for AVC

III. Results and Experiences

PJM OVC system has been run on-line parallel with the existing voltage control system at PJM control center since January 2010. PJM operation engineers performed the study and analysis by comparing the optimal solutions and existing solutions for each interval using software with load flow, contingency assessment and voltage stability assessment. All the assessments are carried out by comparing the existing solutions and the optimal voltage control solutions.

A. Voltage Profiles

The average voltage before and after OVC calculations are compared as shown in Fig 4. The average voltage increases about 0.004 p.u. by OVC.

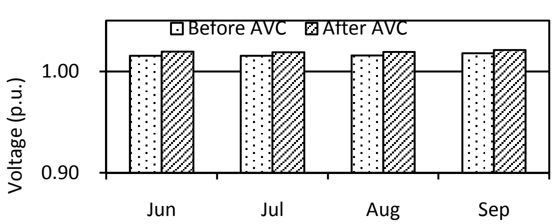

Fig 4 Comparison of Average Voltage Before and After OVC (p.u.)

B. System Losses

The comparisons of average system losses for all the four months are shown as Fig 5. An average loss reduction of about 1.19% for each snapshot can be achieved by OVC.

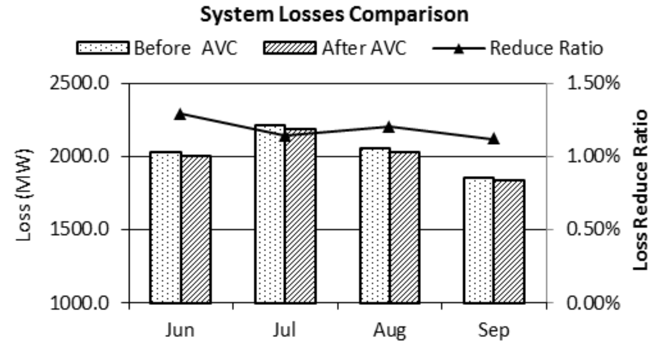

Fig 5 Comparison of Average System Losses Before and After OVC (MW)

C. VAR Reserves

The comparisons of average VAR reserve for all the four months are shown as Fig 6. The average VAR reserve under OVC increased in every month, with an average increase in reserve of 1.08% over four months.

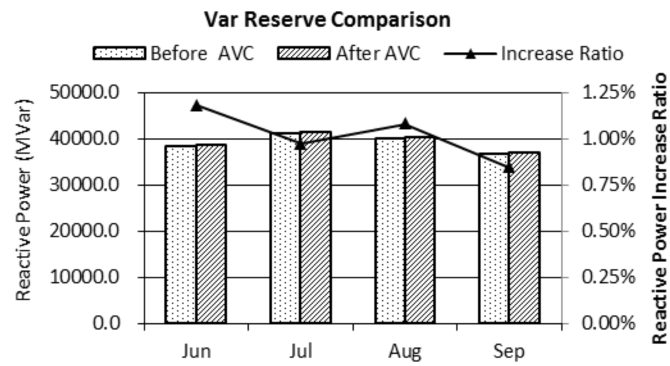

Fig 6 Comparisons of Average VAR Reserve Before and After OVC (MVar)

D. Voltage Violations

1) Base Violations

The comparisons of average base voltage violation numbers for all the four months are shown as Fig 7. An average reduction of 9.3 violations can be achieved by OVC. Nearly 99.3% snapshots were better.

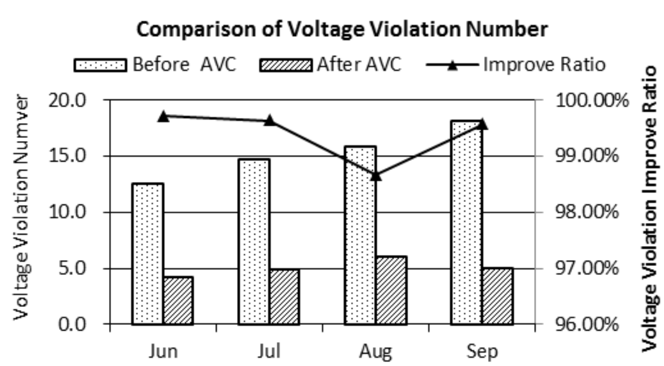

Fig 7 Comparisons of Average Base Voltage Violation Number Before and After AVC

E. Voltage Stability Margins

For evaluating OVC effects on voltage stability margin, the voltage stability margin of five typical interfaces, "EAST", "CENTRAL", "WEST", "BED-BLA" and "APSOUTH", are calculated before and after OVC, shown in Fig 8. The voltage stability margins after OVC are greatly improved. The percentage of increase for the average voltage stability margins are shown in Table 1.

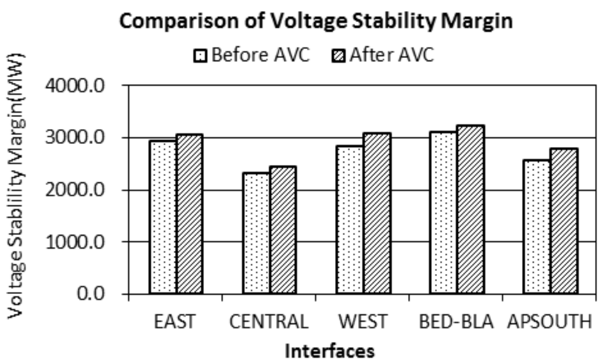

Fig 8 Average Voltage Stability Margins of Five Interfaces (MW)

Table 1. Average Voltage Stability Margins Increase (%)

Time	EAST	CENTRAL	WEST	BED-BLA	APSOUTH
Jun	5.9%	9.0%	12.2 %	6.3%	16.0%
Jul	5.2%	8.2%	12.5%	7.1%	14.4%
Aug	8.5%	9.4%	14.8%	8.4%	14.8%
Sep	10.7%	13.4%	15.8%	8.9%	19.1%

APSOUTH interface is used as example for explaining why the OVC optimal voltage solutions give larger voltage stability margins are greatly improved as following reasons:.

- The more reasonable distribution of reactive power
  The Var reserve in Area13/Zone16 becomes less after OVC, and the voltage becomes higher, that means this area can meet the reactive power needs by itself, and the transmission of reactive power between different areas is reduced
- The more flat voltage distribution
  the mean square deviation of voltage is much smaller after OVC, that means the bus voltages distribution is more flat.
- The higher voltage level
  The bus voltages related to APSOUTH interface become much higher after OVC, especially for receiving side buses

F. Thermal Limits

The influences of AVC on thermal limits are evaluated. The differences of average worst loading before and after AVC are less than 0.3% for all the four months. And more than 98% of the snapshots' worst loading before and after AVC change less than 2%. So AVC's influence on thermal limits is negligible.

IV. Conclusions

The experience of the OVC system running on-line for PJM real time operations has shown the significant potential benefits in PJM system in both aspects of security and economy:

(1) Average reduction of transmission system loss in PJM system is about 1.19%, which means energy savings of more than 220 million kWh/year and monetary savings of more than $17 million/year.
(2) Average increase of system VAR reserve is about 1.08% in PJM.
(3) Improvement of PJM system security for pre-contingency and post-contingency is significant, especially for the voltage stability.

V. References

[1] J.P. Paul, J.T. Leost, and J.M. Tesseron, "Survey of the Secondary Voltage Control in France: Present Realization and Investigations," *IEEE Transactions on Power Systems*, vol. 2, no.2, pp. 505-511, 1987.

[2] M. D. Ilic. Secondary Voltage Control Using Pilot Information. *IEEE Transactions on Power Systems*,vol.3, no.2, pp.660-668, 1988.

[3] P. Lagonotte, J. C. Sabonnadiere, J.Y. Leost, J.P. Paul. Structural analysis of the electrical system: application to secondary voltage control in France. *IEEE Transactions on Power Systems*, vol.4, no.2,pp.479-486, 1989.

[4] V. Arcidiacono, S. Corsi, A. Natale, C. Raffaelli.New Devolopments in the Applications of ENEL Transmission System Automatic Voltage and Reactive Control. *CIGRE*, Report 38/39-06, Paris 1990

[5] S. Corsi, M. Pozzi, C. Sabelli, et al. The coordinated automatic voltage control of the Italian transmission grid - Part I: Reasons of the choice and overview of the consolidated hierarchical system. *IEEE Transactions on Power Systems*,vol.19, no.4,pp.1723-1732, 2004.

[6] S. Corsi, M. Pozzi, M. Sforna, et al. The coordinated automatic voltage control of the Italian transmission grid - Part II: Control apparatuses and field performance of the consolidated hierarchical system. *IEEE Transactions on Power Systems*, vol.19, no.4, pp.1733-1741, 2004.

[7] Hongbin Sun, Qinglai Guo, Boming Zhang, et al. Development and Applications of System-wide Automatic Voltage Control System in China, IEEE PES General Meeting, Calgary, Alberta, Canada, July 26-30, 2009

[8] Qinglai Guo, Hongbin Sun, Jianzhong Tong, et al. Study of System-wide Automatic Voltage Control on PJM System. IEEE PES General Meeting, Minneapolis, Minnesota, USA, July 25-29, 2010

VI. Biographies

Jianzhong Tong(SrM) is a Senior Strategist at PJM Interconnection, he received his BEEE from Zhejiang University, PRC, in 1982, his MSEE from the China Electrical Power Research Institute in 1984, and his Ph.D. from Zhejiang University in 1987. He was an assistant professor at Zhejing University in 1988. He was a visiting scientist at Iowa State University in 1989, and a post doctoral fellow at Cornell University from 1990 to 1993. He was a research and software development engineer at Siemens Power Systems Control from 1993 to 1996. He was a principle engineer at Open Access Technology International, Inc. in 1997. He joined PJM Interconnection, LLC as a senior engineer in 1998.

David W, Souder is the Manager, Operations Planning at PJM Interconnection, he received a Bachelor degree in Electrical Engineering from Drexel University in 1991 and a MBA degree from Villanova University in 1997.

Christopher Pilong received a Bachelor degree in Electrical Engineering from Lehigh University in 2001 and an MBA degree from Villanova

University in 2005. He has been working in PJM real time operation as well as near term operations planning since 2001.

Mingye Zhang received her Bachelor degree from Dept. of Electrical Engineering at Tsinghua University in 2007 and she is now a Ph.D student there. Her research interests include voltage stability analysis and reactive power and voltage optimal control.

Qinglai Guo(M'2008**)** was born in Jilin City, Jilin Province in China on Mar. 6, 1979. He graduated from the Department of Electrical Engineering, Tsinghua University, Beijing, China, in 2000 with MSc. He received his PhD degree from Tsinghua University in 2005 where he is now an associate professor. His special fields of interest include the EMS advanced applications, especially the automatical voltage control.

Hongbin Sun(M'2000) received his double B.S. degrees from Tsinghua University in 1992, the Ph.D from Dept. of E.E., Tsinghua University in 1997. He is now a full professor in Dept. of E.E., Tsinghua Univ, and assistant director of State Key Laboratory of Power Systems in China. From 2007.9 to 2008.9,he was a visiting professor with School of EECS at the Washington State University in Pullman. He is members of IEEE PES CAMS Cascading Failure Task Force and CIGRE C2.13 Task Force on Voltage/Var support in System Operations. His research interests include energy management system, voltage optimization and control, applications of information theory and intelligent technology in power systems. He has implemented system-wide automatic voltage control systems in more than 20 electrical power control centers in China. He won the first rank prize of Beijing science and technology progress in 2004, the first rank award of Chinese national high education achievements in 2005 and the second rank prize of Chinese national technology innovation in 2008 respectively. E-mail: shb@tsinghua.edu.cn

Boming Zhang(Fellow,2009)received his Ph.D. degree from Tsinghua University, Beijing, China, in 1985, in electrical engineering. From 1985 he has progressed from a lecturer to an associate professor and finally to a professor in the Department of Electrical Engineering of Tsinghua University. His research interests include power system analysis and control, especially the EMS advanced applications in EPCC. He is a steering member of CIGRE China State Committee and Int. Workshop of EPCC.